Data Science Fundamentals

Master level

2025/26

Introduction to Data Science

Professor:

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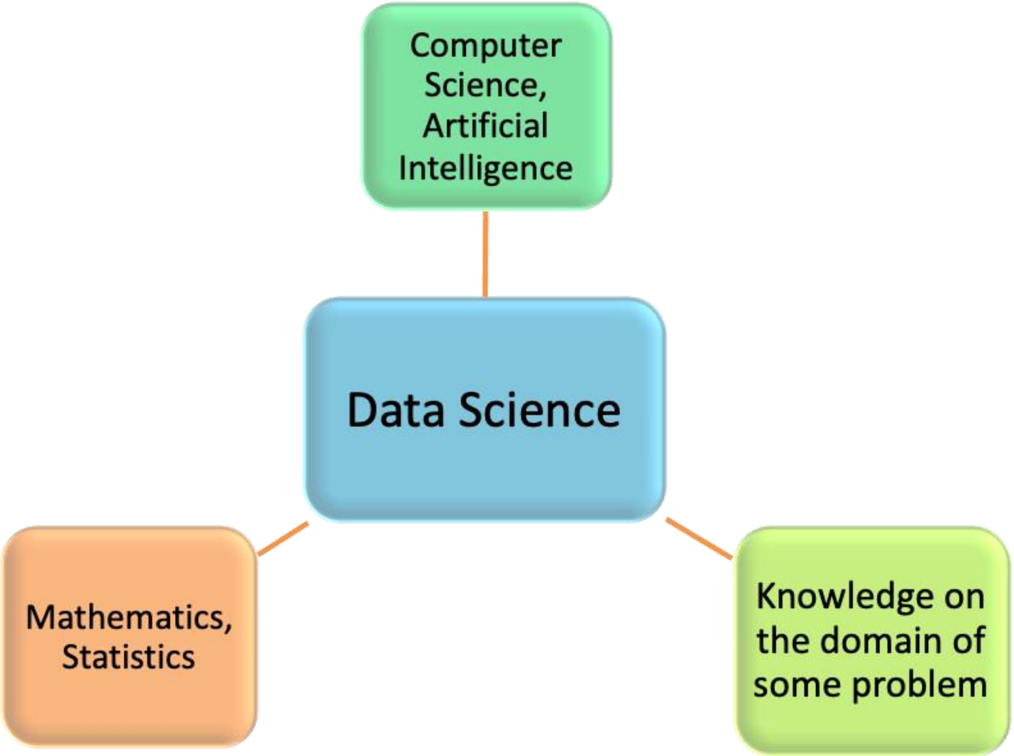
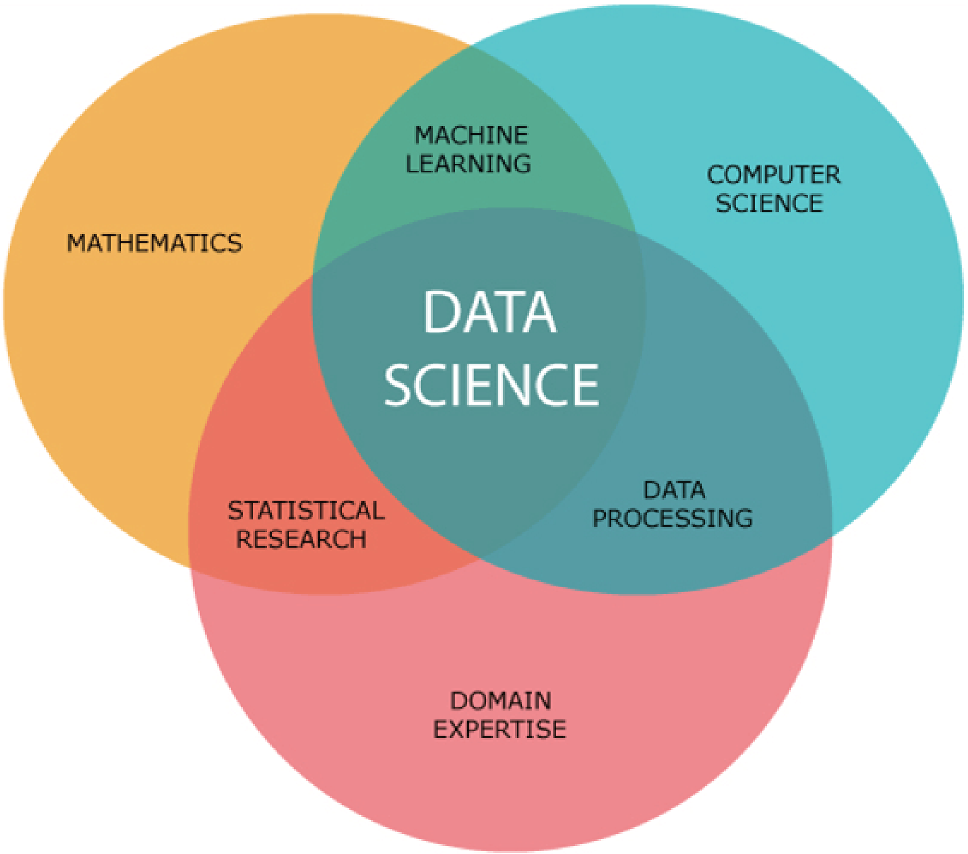
Data Science is a multidisciplinary knowledge domain, applying methods, algorithms, and theories to data in a diversity of formats.

Data Science crosses several scientific domains.

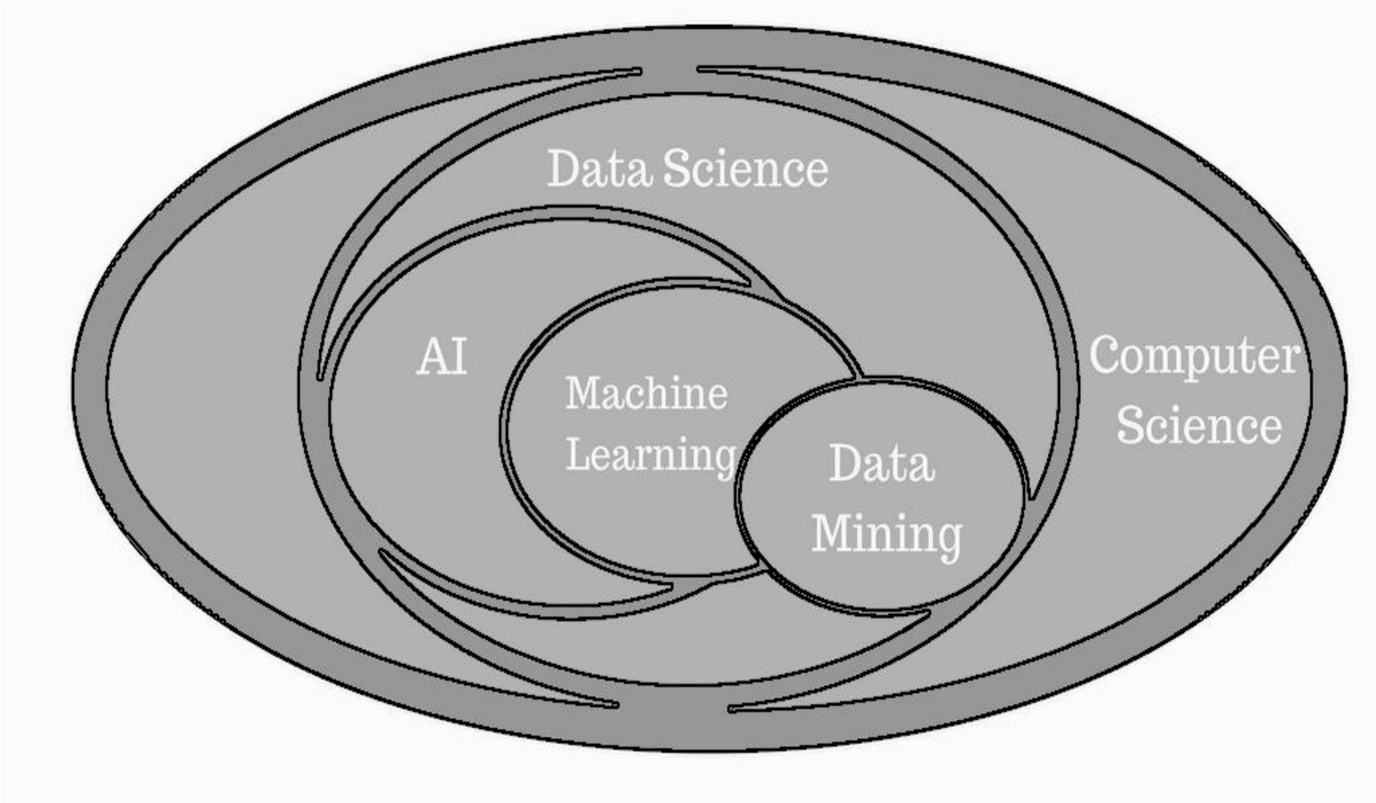


Data Science is a multidisciplinary knowledge domain, applying methods, algorithms, and theories to data in a diversity of formats.

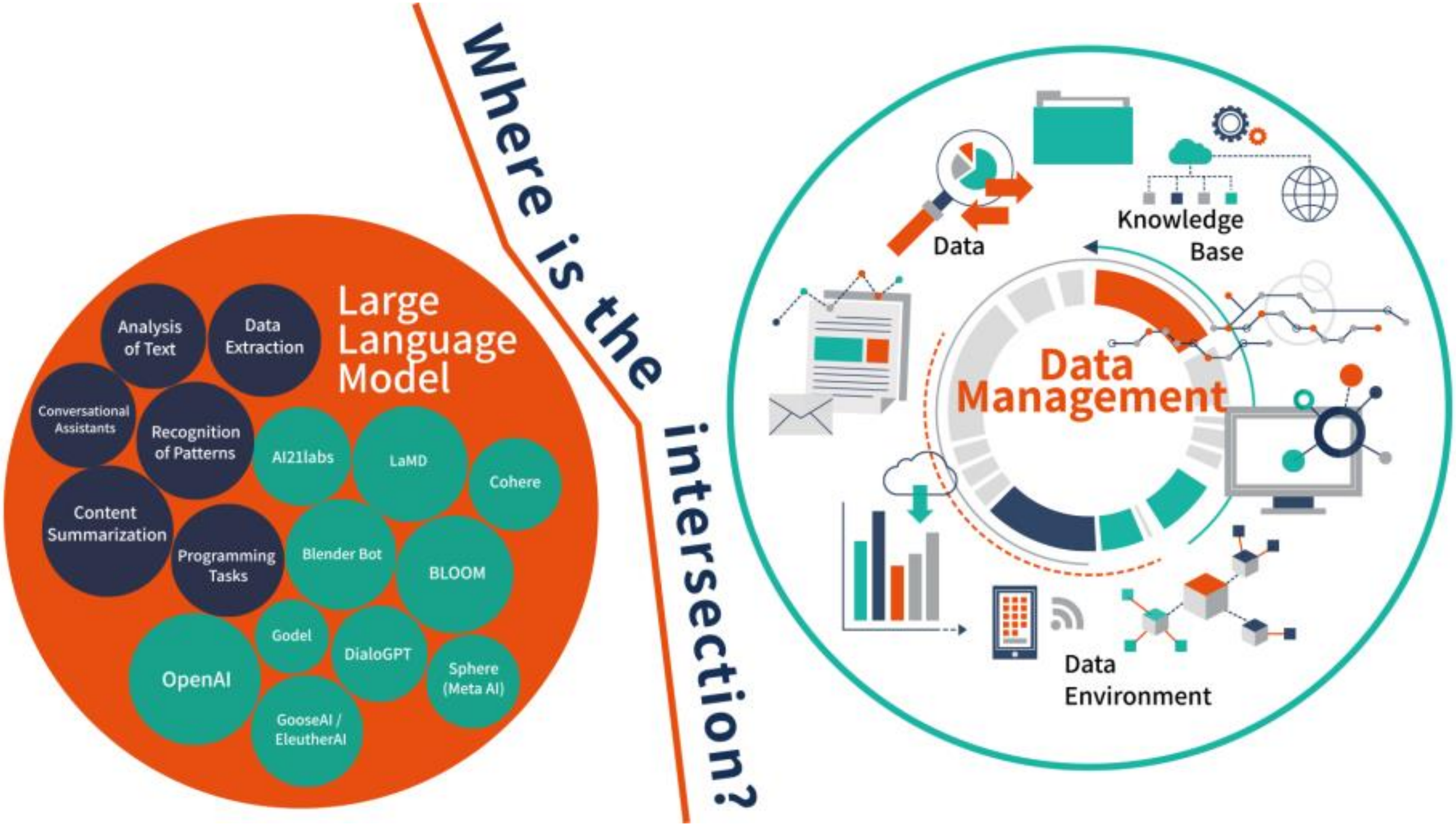
Data Science crosses several scientific domains, as a result of the following sub-domains.



Specifically on the perspective of Computer Science:

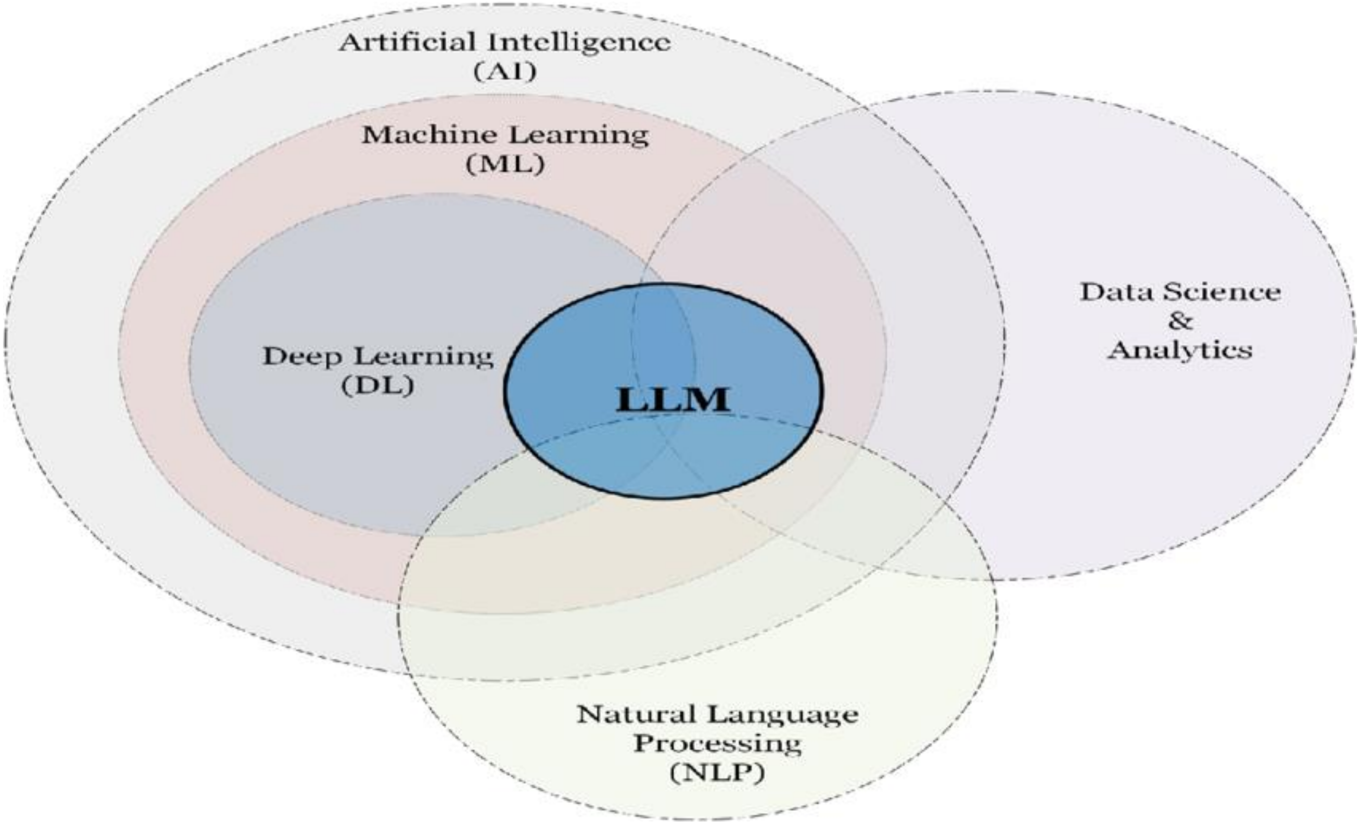


LLMs and Data Science



Attention: LLM falls into “false answers” trap very easily

LLMs and Data Science



The goal is to understand data in a way that allows decision-making. Inferences (how to "complete" and "clean" data) and predictions are needed.

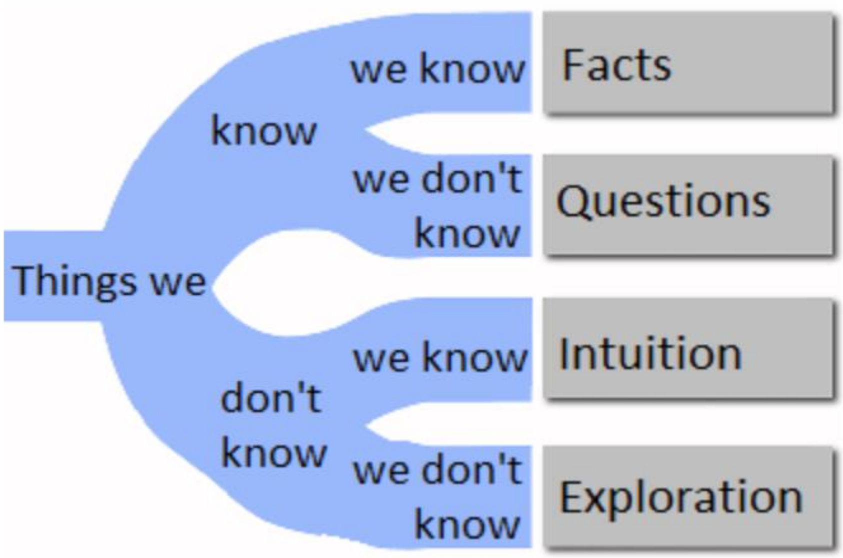
Data analysis is related to past.

Otherwise, Data Science is oriented towards predictions.

Steps

1: Understand and define a problem

A well defined problem is in its way to be solved.



Source: USJournal

Steps

2: Gather and organize data

gather data

collect (gather + organize) data

use collections of data available on the Internet

Steps

3: Quality control and data adjustment

complete vs. incomplete data – the notion of “universe”

data “with noise” vs. data “without noise”

Steps

4: Data exploratory analysis

understand what can be done and what can be measured in data

which are the variables that can be studied?

which statistical aggregates can be measured?

which graphs make sense?

Steps

5: Data modelling

design the model extracted from data – can be represented in a schema, i.e. graphically, or using formulas, or informally described in natural language

the model allows to make predictions

Steps

6: Communication

how to present results

Programme

CP1. Introduction to Data Science

CP2. Introduction to Machine Learning: History, fundamentals, and basic concepts

CP3. Introduction to the Python programming language

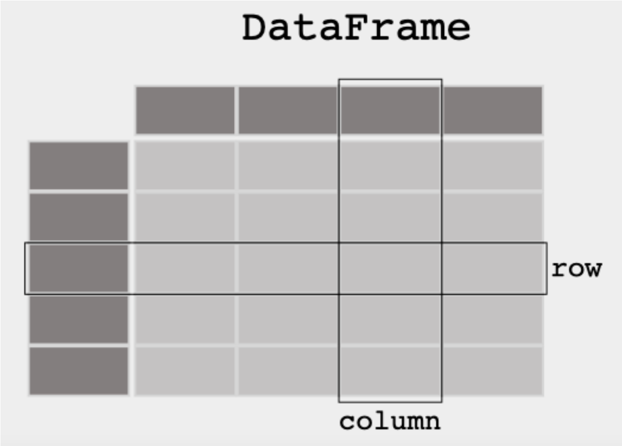
CP4. Exploratory Data Analysis (EDA): Part 1 - Data Wrangling with Pandas

CP5. Exploratory Data Analysis (EDA): Part 2 - Data Visualization with Matplotlib / Seaborn

Programme

CP4. Exploratory Data Analysis (EDA): Part 1 - Data Wrangling with Pandas

" When working with tabular data, such as data stored in spreadsheets or databases, (...) pandas will help you to explore, clean, and process your data. In pandas, a data table is called a DataFrame." in https://pandas.pydata.org/

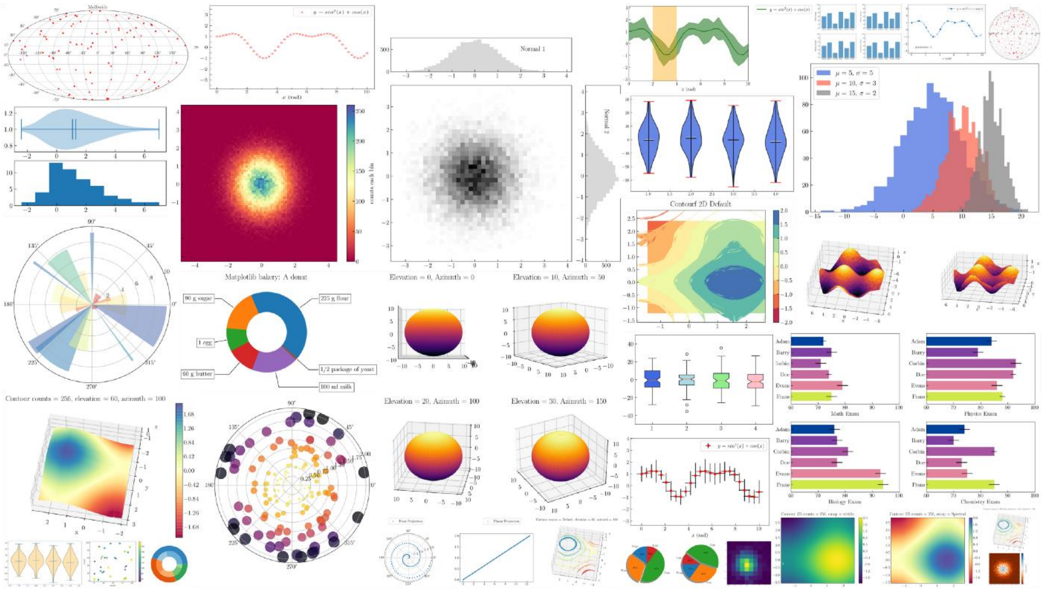


Programme

CP5. Exploratory Data Analysis (EDA): Part 2 - Data Visualization with Matplotlib / Seaborn

"Matplotlib is a (...) library for creating static, animated, and interactive visualizations in Python"

In https://matplotlib.org

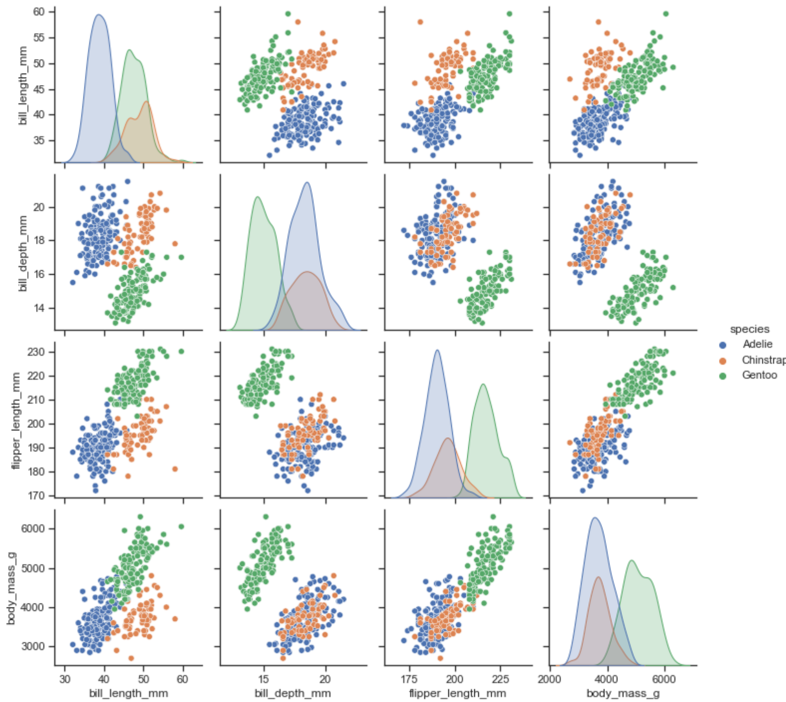


Programme

CP5. Exploratory Data Analysis (EDA): Part 2 - Data Visualization with Matplotlib / Seaborn

"Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing (...) statistical graphics."

In https://seaborn.pydata.org



Programme

CP6. Supervised Learning: SVM, Decision Trees, Linear and Logistic Regressions, Random Forests

The algorithm is trained on a set of data inputs and outputs, with a goal of learning the structure in data.

CP7. Unsupervised Learning: K-means clustering; Reinforcement Learning: Q-Learning

The learning algorithm works to discover the structure in the input on its own.

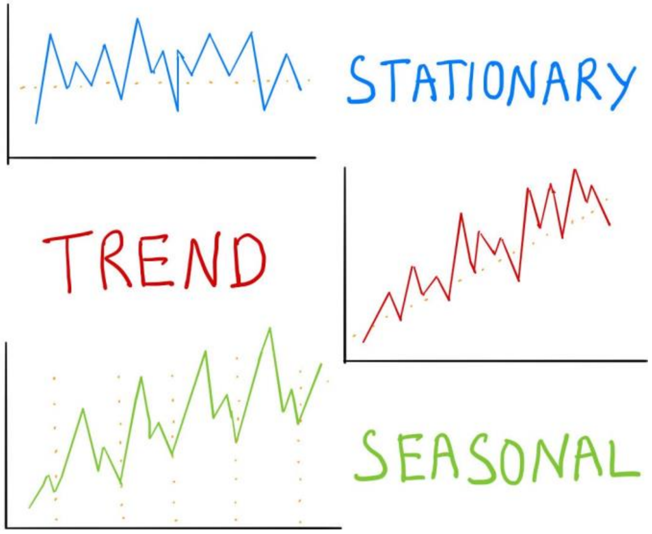
CP8. Classification and Regression Problems; Numeric / Continuous and Categorical / Discrete Variables

CP9. Artificial Neural Networks

Programme

CP10. Deep Learning: Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs); Computer Vision

CP11. Time Series



In https://medium.com/@namnguyenthe

Assessment

Given the nature of the contents, the evaluation will encompass a project.

Its subject should be aligned with all or part of the syllabus.

Project subject proposal (5%).

Project (95%, including teamwork (report and software), and oral exam).

Presence in class is not mandatory.

There is no final exam.

Example

Google Flu Trends

In 2008 Google proposed to map in real time the flu outbreaks using the search related to flu in its search engine.

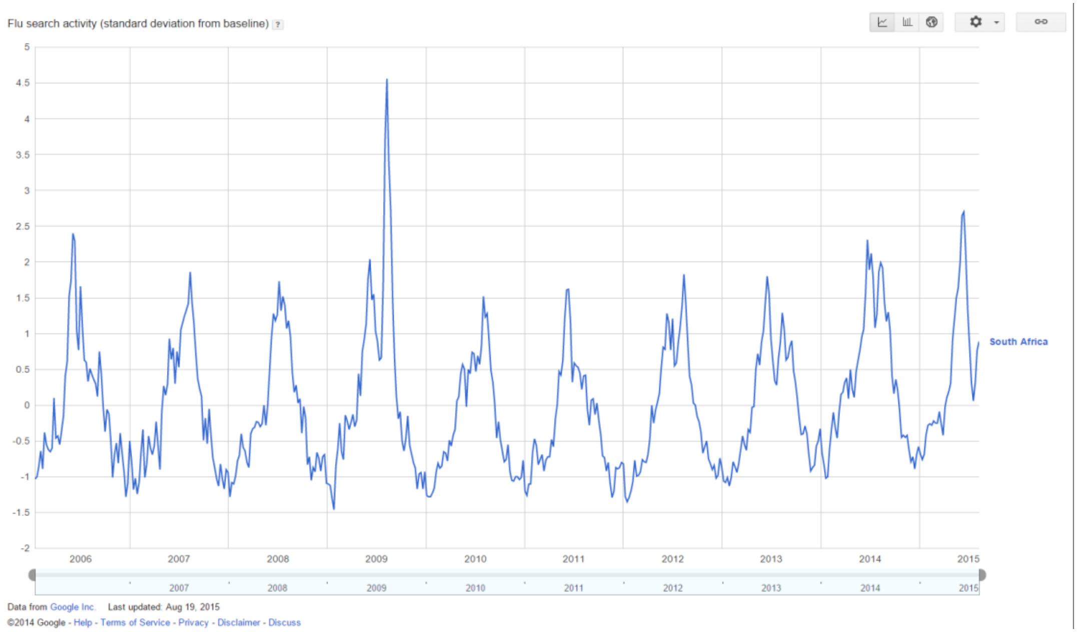
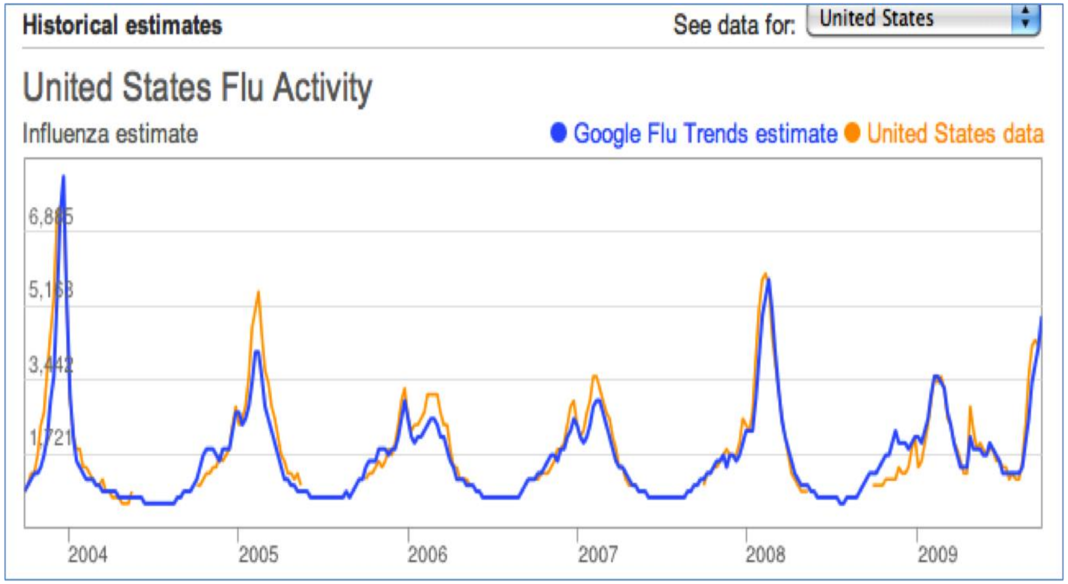


Image: data from South Africa

Example

Google Flu Trends

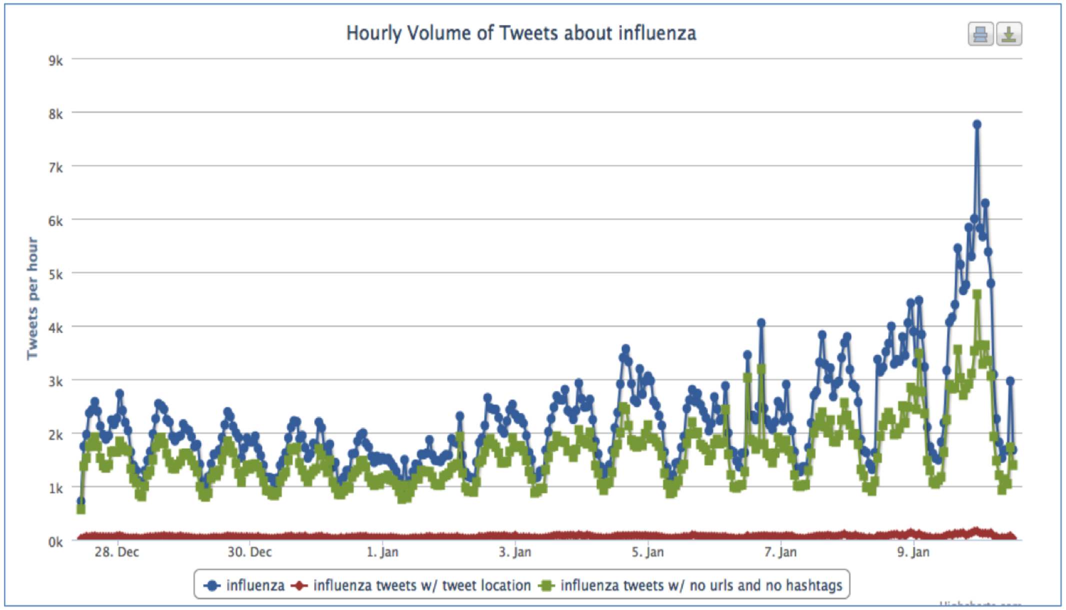
Results were interesting, but not exact.



Example

Google Flu Trends

Google Flu Trends inspired several developments, including Mappyhealth that monitors the propagation of 27 different types of virus from tweets on Twitter.



Example

Forecasting Seasonal Influenza Fusing Digital Indicators and a Mechanistic Disease Model

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ABSTRACT

The availability of novel digital data streams that can be used as proxy for monitoring infectious disease incidence is ushering in a new era for real-time forecast approaches to disease spreading. Here, we propose the first seasonal influenza forecast framework based on a stochastic, spatially structured mechanistic model (individual level microsimulation) initialized with geo-localized microblogging data. The framework provides for more than 600 census areas in the United States, Italy and Spain, the initial conditions for a stochastic epidemic computational model that generates an ensemble of forecasts for the main indicators of the epidemic season: peak time and intensity. We evaluate the forecasts accuracy and reliability by comparing the results with the data from the official influenza surveillance systems in the US, Italy and Spain in the seasons 2014/15 and 2015/16. In all countries studied, the proposed framework provides reliable results with leads of up to 6 weeks that became more stable and accurate with progression of the season. The results for the United States have been generated in real-time in the context of the Centers for Disease Control and Prevention "Forecasting the Influenza Season Challenge". A characteristic feature of the mechanistic modeling approach is in the explicit estimate of key epidemiological parameters relevant for public health decision-making that cannot be achieved with statistical models that do not consider the disease dynamic. Furthermore, the presented framework allows the fusion of multiple data streams in the initialization stage and can be enriched with census, weather and socioeconomic data.

Keywords

Computational epidemiology; Influenza modeling; Real-time forecasting; Social media

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1. INTRODUCTION

Seasonal influenza annually results in up to 5 million severe illness, half million deaths, and increases visits to emergency departments [1, 2, 3, 4]. Real-time forecast of major influenza indicators, such as peak time, and peak intensity can provide key information for public health interventions, such as resources allocation for influenza prevention, control and the public communication of health risk. Although substantial work has been carried out in the field of infectious disease analysis and modeling in the past, real-time forecasting is now fueled by the availability of novel digital data streams generated by human activities that can provide real-time surrogates for the clinically-based reporting of influenza-like-ilness (ILI) [5, 6]. Typical examples are provided by Google Flu Trends (GFT) and similar approaches based on Twitter, Wikipedia and other datasets originated by human activities in the digital world. Combined with traditional surveillance these novel data streams are being used to provide local and timely information about disease and health indicators in populations around the world [7, 8]. Although novel digital data streams may suffer from a number of limitations including signal drifts that might affect the reliability of their forecasts [9, 10], they have increasingly large data volumes, are highly contextual, geo-localized, and allows an unprecedented real-time access to information that can improve forecasting methodologies.

Novel data streams have been generally used as input for statistical models that do not take into account a detailed individual level description of the disease dynamics. Here, we introduce a general framework that, fusing digital indicators with a stochastic, spatially structured, individual level, disease dynamic model, produces short and long term predictions of seasonal influenza at different granularities. In particular, we use geo-localized microblogging data from Twitter to quantify relative geographical incidence of influenza in a given country. These estimates serve as inputs for a mechanistic ILI dynamic model: the global epidemic and mobility model (GLEAM) [11, 12, 13]. We explore the disease dynamic by a latin hypercube sampling of the model's parameter and initial conditions space. For each sampled point we generate 500 identically initialized Monte Carlo simulations of the epidemic spreading that can be aggregated at different geographical resolutions. Finally, we perform a likelihood analysis assimilating official ILI surveillance data up to the time of forecast, thus selecting the ensemble of numer-

Forecasting Seasonal Influenza Fusing Digital Indicators and a Mechanistic Disease Model

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Exemplos

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COMMENTARY

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The future of influenza forecasts

Cécile Viboud[a,1] and Alessandro Vespignani[b,c,d]

Recent years have seen a growing interest in generating real-time epidemic forecasts to help control infectious diseases, prompted by a succession of global and regional outbreaks. Increased availability of epidemiological data and novel digital data streams such as search engine queries and social media (1, 2), together with the rise of machine learning and sophisticated statistical approaches, have injected new blood into the science of outbreak forecasts (3, 4). In parallel, mechanistic transmission models have benefited from computational advances and extensive data on the mobility and sociodemographic structure of human populations (5, 6). In this rapidly advancing research landscape, modeling consortiums have generated systematic model comparisons of the impact of new interventions and ensemble predictions of outbreak trajectory, for use by decision makers (7–12). Despite the rapid development of disease forecasting as a discipline, however, and the interest of public health policy makers in making better use of analytics tools to control outbreaks, forecasts are rarely operational in the same way that

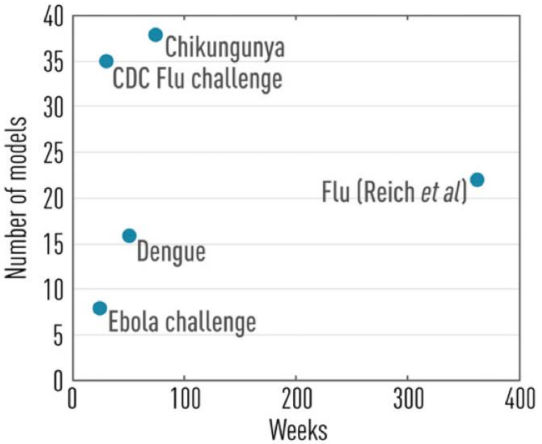
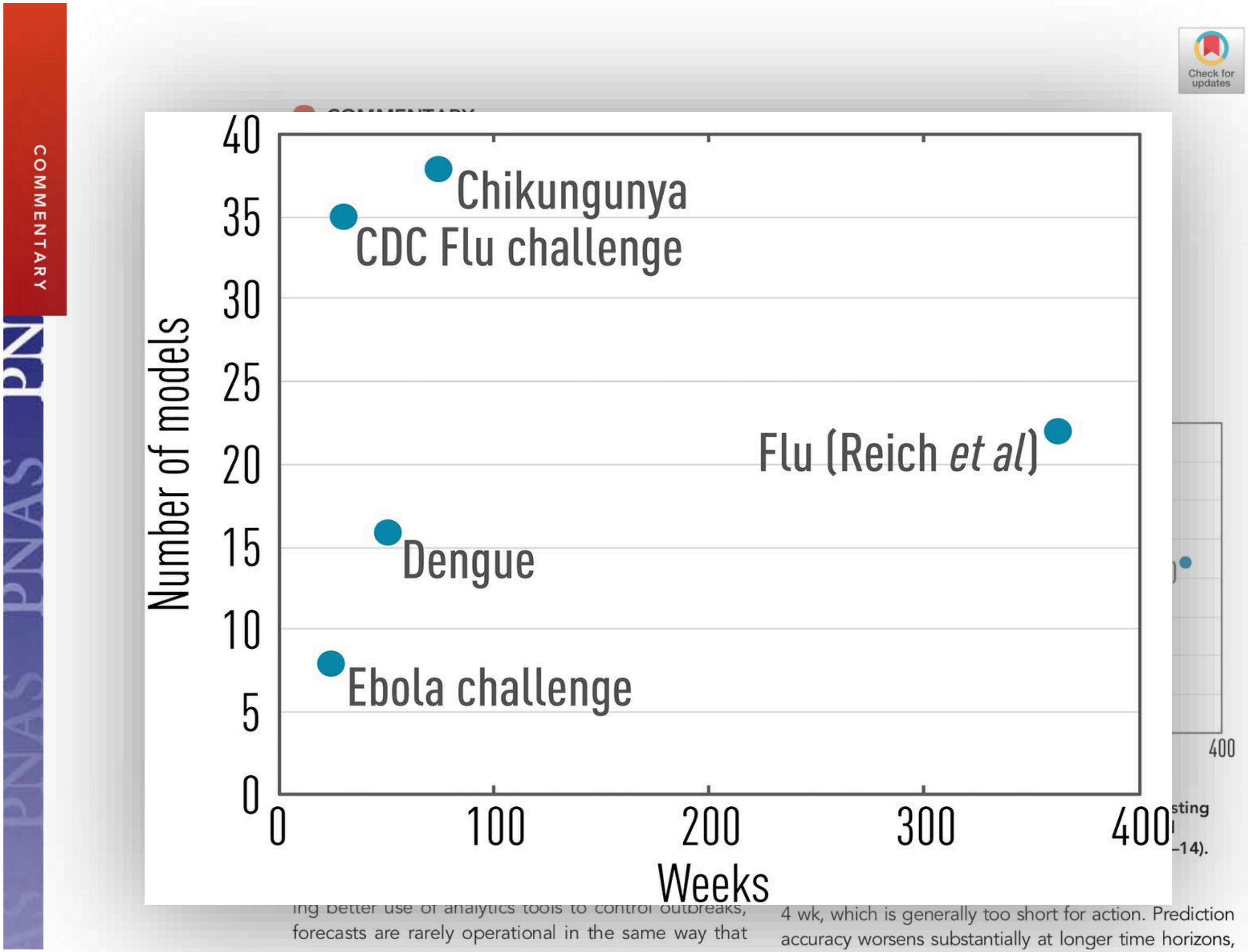


Fig. 1. Past and present infectious disease forecasting challenges as a function of prediction horizon and number of models considered (data from refs. 10–14).

4 wk, which is generally too short for action. Prediction accuracy worsens substantially at longer time horizons,

In: Proceedings of the National Academy of Sciences



In: Proceedings of the National Academy of Sciences